

Purchasing-Issuing OPD tickets in a Government Hospital: A case study on Queueing Process Analysis

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Introduction

- **Framework:** Queueing theory applies stochastic models and probability to study waiting lines, evaluating critical metrics such as arrival pattern, server utilization, and system pattern.
- **Importance In Healthcare:** Outpatient Department (OPD) ticket counters experience dynamic, fluctuating traffic workflows. Managing long queues is crucial to maintaining operational throughput and patient satisfaction.
- **Fieldwork Objective:** Direct continuous observation was conducted at Barasat Government Medical College and Hospital to model patient flows under empirical arrival and service constraints.
- **Focus:** Examined real-world departures from classical Markovian assumptions (constant arrival rates λ and service rate μ) to establish an advanced, customized queueing model.

Data Collection Overview

- Primary data was collected via direct, continuous observation at the OPD ticket counters of Barasat Government Medical College and Hospital to track real-time queueing behavior. The study monitors a high-volume, multi-server operational framework operating across two distinct registration ticket counters.
- **Patient Arrival :** Collected 690 precise arrival to compute consecutive inter-arrival durations and analyze the time-dependent arrival rate ($\lambda_n(t)$).
- **Service Completion:** Collected 263 individual service entry and completion points ("Service Time IN" / "Service Time OUT") to mathematically evaluate the state-dependent service rate (μ_n).
- **Queue Dynamics:** The physical queue lengths across two operational counters (Queue 1 and Queue 2), along with the concurrent absolute system length (n).

Demo Dataset

- Patient Arrival and Inter-Arrival Time Dataset Sample

No. of Patients	Arrival Time	Inter Arrival Time (in min.)
1	8:37:28	0
2	8:37:49	0.35
3	8:38:23	0.566666667
4	8:40:11	1.8
5	8:40:25	0.233333333
6	8:40:50	0.416666667
7	8:41:10	0.333333333
8	8:41:25	0.25
9	8:42:27	1.033333333
10	8:42:40	0.216666667
11	8:42:55	0.25
12	8:43:12	0.283333333
13	8:43:20	0.133333333
14	8:43:28	0.133333333
15	8:43:36	0.133333333

Table: Sample Dataset of patient arrivals and calculated inter-arrival times. 🔍 🔗 🔍

Demo Dataset

- Patient Service Dataset Sample :

No. of Patients	Service Time (IN)	Service Time (OUT)
1	8:35:00	8:38:00
2	8:36:00	8:39:00
3	8:38:00	8:39:00
4	8:38:00	8:40:00
5	8:39:00	8:41:00
6	8:39:00	8:42:00
7	8:40:00	8:42:00
8	8:40:00	8:42:00
9	8:42:00	8:44:00
10	8:42:00	8:43:00
11	8:43:00	8:44:00
12	8:42:00	8:44:00
13	8:44:00	8:45:00
14	8:44:00	8:45:00
15	8:45:00	8:46:00

Table: Sample dataset patient counter service durations. < ≡ ≡ ≡

Demo Dataset

- Multi-Queue Length Status Sample

Time	Queue 1	Queue 2	System Length(n)	Total Queue Length
8:35	1	3	10	4
8:45	1	3	15	4
8:50	3	11	14	14
9:00	3	5	25	8
9:10	6	9	30	15
9:20	10	10	40	20
9:25	8	10	24	18
9:30	8	16	28	24
9:35	13	27	45	40
9:40	14	33	60	47

Table: Dataset of system tracking state parameters and line accumulation points.

Description of Poisson Process

- **Independent Increments:** Arrivals in non-overlapping intervals are statistically independent; past arrival patterns exert zero influence on future traffic.
- **Stationary Increments:** Arrival probabilities within any window depend strictly on the absolute length of the interval, rather than its specific starting epoch.
- **Rare Simultaneous Events:** The likelihood of experiencing multiple arrivals at the exact same split second (Δt) is asymptotically zero.

Notations and System Variables

- **Number of Arrivals** $X(t)$: Counting process tracking the number of patient arrivals in the system from baseline (time 0) up to time t .
- **Number of Departures** $Y(t)$: Counting process tracking the total number of counter service completions (departures) that have occurred up to time t .
- **Number of Services** $N(t)$: Number of Services takers in the system (here in Hospital) at time t .
- **Arrival Rate** $\lambda_n(t)$: Expected number of arrivals per unit time given the system is exactly in state n at time t . State-dependence accommodates complex behaviors like customer balking.
- **Service Rate** $\mu_n(t)$: Expected number of departures per unit time when the system holds n customers at time t . It remains static (μ) for single servers or scales dynamically ($n\mu$ or $c\mu$) in multi-server configurations.

Assumptions of Poisson Process

- **Counting Framework:** Let $X(t)$ denote the total number of arrivals observed up to time t . The increment $X(t + \Delta t) - X(t)$ represents the number of arrivals occurring during the small time interval $(t, t + \Delta t]$.

- **Probability of a Single Arrival:** For a sufficiently small interval Δt ,

$$P(X(t + \Delta t) - X(t) = 1) = \lambda \Delta t + o(\Delta t)$$

That is, the probability of exactly one arrival is proportional to the interval length.

- **Probability of Multiple Arrivals:** For a sufficiently small interval t

$$P(X(t + \Delta t) - X(t) \geq 2) = o(\Delta t)$$

Thus, the probability of two or more arrivals is negligible.

- **Independent increments:** The number of arrivals in one interval is independent of any other non-overlapping interval.
- **Stationary increments:** The distribution of arrivals depends only on the length of the interval and not on its position in time.

Plot of Arrival Rate Against Time Intervals

- To analyze how the arrival process changes over time at the OPD ticket counters, the arrival rate $\lambda(t)$ is plotted against specific 5-minute time intervals.

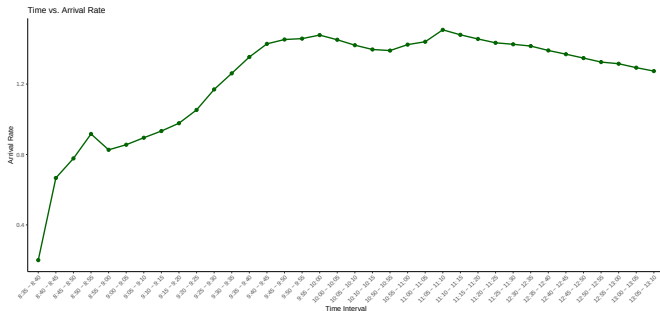


Figure: Time Interval vs. Arrival Rate $\lambda(t)$

- We observe the early morning opening rush at 8:35–8:40 AM through mid-morning peaks, and continuing until the afternoon cool-down periods ending around 13:05–13:10 PM. So, it's time dependent.

To Fit a Polynomial

- Here our curve is of concave type. So, by fitting a 2nd-degree polynomial curve to all the data points:

$$\lambda(t) = \beta_0 + \beta_1 t + \beta_2 t^2$$

- Empirical Parameter Solutions:** Direct estimation from the dataset yields the unique parameters:

$$\beta_0 = 1.1973677394, \beta_1 = 0.0551898023, \beta_2 = -0.000232203$$

- Then the final analytical function for the arrival process:

$$\lambda(t) = 1.1973677394 + 0.0551898023 t - 0.0002322031 t^2$$

Distribution of Inter-Arrival Time

Let us assume that throughout the time duration of length t , the inter-arrival time follows approximately an exponential distribution with mean $\frac{1}{\lambda(t)}$.

As a result, we can infer that the corresponding arrival process $X(t)$ is a time-dependent Poisson process, where

$$X(t) \sim \text{Poisson}(m(t))$$

where,

$$m(t) = \int_0^t \lambda(x) dx \quad (1)$$

A key result from queueing theory states that **if inter-arrival times are independent and identically distributed and follow an Exponential distribution with parameter λ , then the counting process follows a Poisson process with mean λt .**

Reference: J. Medhi, *Stochastic Models in Queueing Theory*, 2nd ed., Academic Press, 2002, p. 150.

Distribution of Inert-Arrival Time

Also we have found that over the entire time interval of study out of 3 selected time distributions exponential, gamma, weibull. here exponential fits well.

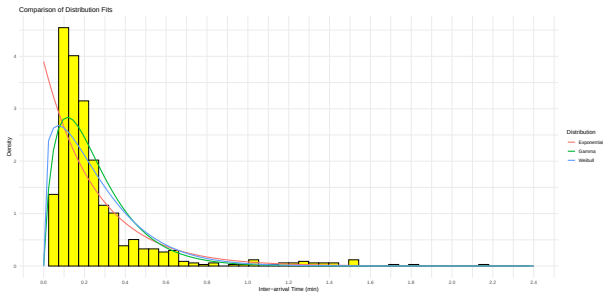


Figure: Empirical Inter-Arrival Density vs. Parametric Fits

AIC and BIC Comparison

Now we use AIC and BIC to choose the best model. For both of these, a lower value means a better fit.

Distribution	AIC (Akaike Info Criterion)	BIC (Bayesian Info Criterion)
Exponential	-683.66	-674.6
Gamma	-664.05	-650.45
Weibull	-658.12	-644.52

- From the comparison table, the Exponential distribution has the lowest AIC and BIC values. Hence, the Exponential distribution is selected as the best-fitting model for inter-arrival times.
- We can treat this discussion as a direct or indirect evidence in our assumptions of exponential distribution for our inter arrival time distribution.

Time-Dependent Poisson Process (Non-Homogeneous Poisson Process)

To accurately capture fluctuations across a day, we transition to a Time-Dependent Poisson Process (also known as a Non-Homogeneous Poisson Process). Under this framework, the arrival rate is treated as a non-random function of time, denoted by $\lambda(t)$.

Theorem: The probability generating function (p.g.f.) of a non-homogeneous Poisson process $\{X(t), t \geq 0\}$ is given by:

$$Q(s, t) = \exp\{m(t)(s - 1)\},$$

where

$$m(t) = \int_0^t \lambda(x) dx$$

is the expected value of $N(t)$.

Source: J. Medhi, *Stochastic Models in Queueing Theory*, 2nd ed., Academic Press, 2002, p. 161.

Probability Mass Function

In the previous section, we derived the analytical form of the time-dependent arrival rate function $\lambda(t)$. Integrating $\lambda(t)$ over the interval $[0, t]$, we obtain the estimated cumulative intensity function $m(t)$. is

$$m(t) = 1.1973677394 t + 0.02759490115 t^2 - 0.0000774010333 t^3$$

Substituting this cumulative intensity polynomial into the general distribution expression of the final arrival state probability model:

$$p_k(t) = \frac{[m(t)]^k e^{-m(t)}}{k!}$$

Plot of Service Rate Against Time Intervals

- To analyze how the service departure process changes over time at the OPD ticket counters, the service rate μ is plotted against specific 5-minute time intervals.

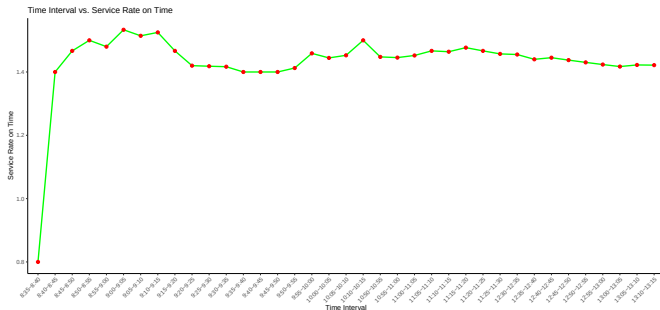


Figure: Time Interval vs. Arrival Rate $\lambda(t)$

- We observe the constant behavior of the service rate around all the 5-minutes time intervals. So, there is no time dependency.

Distribution of Inter-Service Time

- To find the distribution of inter-service times. We draw the histogram. And, we are fitted Exponential, Gamma, Erlang-6 Erlang-3 and Weibull distributions.

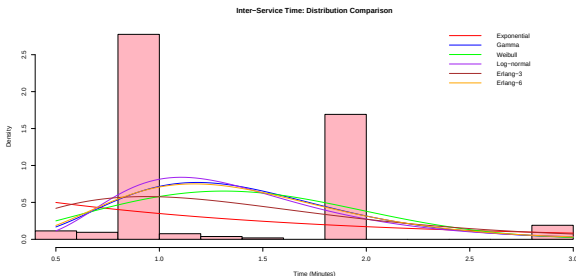


Figure: Empirical Inter-Arrival Density vs. Parametric Fits

- Now we formulate the goodness of fit test. So we use Kolmogorov-Smirnov test.

Distribution of Inter-Service Time

- To establish the goodness of fit criteria we want to test
Null hypothesis(H_0): the data follows the expected distribution
Alternative hypothesis(H_1): the data does not follow the expected distribution.
Hence in our case the p-value of Erlang-6, Erlang-3, Gamma, Weibull, Exponential distributions all are less than 0.05. So, we reject the null hypothesis. And hence we could not get any information by formulating this test so focus on AIC and BIC values.
- Therefore we use the AIC criterion to find the fitted distribution among the selected distributions and the resulted values are

Distribution of Inter-Service Time

Table: AIC Comparison of Candidate Distributions

Rank	Distribution	AIC
1	Erlang-3	418.0909
2	Gamma	422.49
3	Weibull	442.38
4	Exponential	706.01

- From the above table we may conclude that the Erlang-3 distribution is better fit the distribution of inter-arrival times with respect to the other distributions.

Distribution of Service Completions

- If the individual service completion times follow an Erlang- k distribution, its probability density function (p.d.f.) is defined as:

$$f(t) = \frac{\mu^k}{\Gamma(k)} e^{-\mu t} t^{k-1}, \quad t > 0; \quad k > 0, \quad \mu > 0 \quad (2)$$

- This expression is identical to a Gamma(μ, k) distribution, so expected service completion time equal to $\frac{k}{\mu}$.
- If $S_n = T_1 + T_2 + \cdots + T_n =$ time taken for n service completion. Where, $T_i \sim \text{Gamma}(\mu, k)$ independently, then by the additive property of the Gamma distribution,

$$S_n \sim \text{Gamma}(\mu, nk) \quad \text{and} \quad S_{n+1} \sim \text{Gamma}(\mu, (n+1)k) \quad (3)$$

- Let, $Y(t) =$ no. of service completion during a time interval of length t unit.

Distribution of Service Completions

- Then,

$$P[Y(t) = n] = P[S_n \leq t] - P[S_{n+1} \leq t] \quad (4)$$

$$\begin{aligned} P[Y(t) = n] &= \int_0^t \frac{\mu^{nk}}{\Gamma(nk)} e^{-\mu u} u^{nk-1} du - \int_0^t \frac{\mu^{(n+1)k}}{\Gamma((n+1)k)} e^{-\mu u} u^{(n+1)k-1} du \\ &= I_{\mu t}(nk) - I_{\mu t}(\overline{n+1}k) \end{aligned} \quad (5)$$

- where $I_x(n) = \int_0^x \frac{e^{-u} u^{n-1}}{\Gamma(n)} du$ represents the incomplete Gamma integral.
- If a discrete random variable $X \sim \text{Poisson}(\lambda)$, then,

$$P[X \leq n] = 1 - I_\lambda(n+1) \quad (6)$$

Using this identity, the functional difference $I_{\mu t}(nk) - I_{\mu t}(\overline{n+1}k)$ can be rewritten as:

$$I_{\mu t}(nk) - I_{\mu t}(\overline{n+1}k) = \{1 - I_{\mu t}(\overline{n+1}k)\} - \{1 - I_{\mu t}(nk)\} \quad (7)$$

Distribution of Service Completions

$$\begin{aligned} &= P[X \leq nk + k - 1] - P[X \leq nk - 1], \quad \text{where } X \sim \text{Poisson}(\mu t) \\ &= P[nk \leq X \leq nk + k - 1] \\ &= \sum_{x=nk}^{nk+k-1} \frac{e^{-\mu t} (\mu t)^x}{x!} \end{aligned} \quad (8)$$

So,

$$P[Y(t) = n] = \sum_{x=nk}^{nk+k-1} \frac{e^{-\mu t} (\mu t)^x}{x!}, \quad n = 0, 1, 2, \dots \quad (9)$$

where k is a known positive integer. In our case, as we work with the Erlang-3 distribution as the distribution of inter-service completion times, we have found out and established that **the counting distribution of $Y(t)$ follows a 3-Erlang Renewal Count Process**. Now,

$$E(Y(t)) = \sum_{n=0}^{\infty} nP[Y(t) = n] = \sum_{n=0}^{\infty} n\{P[S_n \leq t] - P[S_{n+1} \leq t]\} \quad (10)$$

Distribution of Service Completions

$$\begin{aligned} &= (1 \cdot P[S_1 \leq t] - 1 \cdot P[S_2 \leq t]) + (2 \cdot P[S_2 \leq t] - 2 \cdot P[S_3 \leq t]) \\ &\quad + (3 \cdot P[S_3 \leq t] - 3 \cdot P[S_4 \leq t]) + \dots \\ &= P[S_1 \leq t] + P[S_2 \leq t] + P[S_3 \leq t] + \dots \end{aligned} \quad (12)$$

$$\begin{aligned} &= I_{\mu t}(k) + I_{\mu t}(2k) + I_{\mu t}(3k) + \dots \\ &= P[X > k - 1] + P[X > 2k - 1] + P[X > 3k - 1] + \dots \end{aligned} \quad (13)$$

Alternatively,

$$E(Y(t)) = \sum_{n=1}^{\infty} n \cdot \sum_{x=nk}^{nk+k-1} \frac{e^{-\mu t} (\mu t)^x}{x!} \quad (14)$$

By changing the order of summation, bounding the inner parameter to $nk \leq x \leq (n+1)k - 1$, which simplifies to the floor function limit $n \leq \frac{x}{k}$ and $n \geq \frac{x}{k}$:

Distribution of Service Completions

$$\begin{aligned} &= e^{-\mu t} \sum_{x=k}^{\infty} \frac{(\mu t)^x}{x!} \sum_{n \geq \lceil x/k \rceil}^{\infty} n \\ &= e^{-\mu t} \sum_{x=k}^{\infty} \left\lfloor \frac{x}{k} \right\rfloor \frac{(\mu t)^x}{x!} \end{aligned} \quad (15)$$

$$= e^{-\mu t} \left\{ 1 \cdot \left[\frac{(\mu t)^k}{k!} + \dots + \frac{(\mu t)^{2k-1}}{(2k-1)!} \right] + 2 \cdot \left[\frac{(\mu t)^{2k}}{2k!} + \dots + \frac{(\mu t)^{3k-1}}{(3k-1)!} \right] + \dots \right\} \quad (16)$$

This expression has no definite, closed-form solution for general values of k . From different sources it can be approximate it as,

$$\approx \frac{\mu t}{k} \quad (17)$$

Distribution of Service Completions

To evaluate the probability of exactly one completion over a small time increment $\Delta t \rightarrow 0$, we set $t = \Delta t$ for a single event ($n = 1$):

$$P[Y(\Delta t) = 1] = \sum_{x=k}^{2k-1} \frac{e^{-\mu\Delta t}(\mu\Delta t)^x}{x!} = e^{-\mu\Delta t} \left[\frac{(\mu\Delta t)^k}{k!} + \frac{(\mu\Delta t)^{k+1}}{(k+1)!} + \dots \right] \quad (18)$$

$$= \left\{ 1 - \mu\Delta t + \frac{(\mu\Delta t)^2}{2!} - \dots \right\} \left[\frac{(\mu\Delta t)^k}{k!} + \frac{(\mu\Delta t)^{k+1}}{(k+1)!} + \dots \right] \quad (19)$$

$$P[Y(\Delta t) = 1] \approx \frac{(\mu\Delta t)^k}{k!} \approx \frac{\mu\Delta t}{k} \quad (20)$$

Queueing Structure of the OPD Ticketing System

Queueing Model: According to Kendall's queueing notation, the OPD ticketing system is represented as

$$M_t/E_3/2/FCFS/\infty/$$

- M_t (**Arrival Process**): Time-dependent Poisson process with varying arrival rate $\lambda(t)$.
- E_3 (**Service Process**): Service times follow an Erlang-3 distribution.
- **2 (Servers)**: Two OPD ticket counters operate in parallel.
- ∞ (**System Capacity**): Indicates infinite system capacity.
- **FCFS (Queue Discipline)**: Patients are served on a First-Come, First-Served basis.

Definitions and Differential Difference Formulations for the Queueing Model

Let $N(t)$ denote the random variable representing the total number of patients present in the hospital OPD system at time t .

For the Erlang- k service distribution setup, the effective service completion rate function μ_n when the system occupies state $N(t) = n$ is defined as:

$$\mu_n = n \left(\frac{\mu}{k} \right), \quad 1 \leq n < c \quad (21)$$

$$\mu_n = c \left(\frac{\mu}{k} \right), \quad n \geq c \quad (22)$$

As in our case time-dependent arrival intensity $\lambda(t)$ be a concave function of time t having a λ_{max} and λ_{min} within the active observation case. We analyze the system state changes across a small time increment $\Delta t \rightarrow 0$.

Definitions and Differential Difference Formulations for the Queueing Model

So,

$$P_0(t + \Delta t) = P_0(t) \{1 - \lambda(t) \Delta t + o(\Delta t)\} + P_1(t) \left\{ \left(\frac{\mu}{k} \right) \Delta t + o(\Delta t) \right\} \quad (23)$$

Then,

$$\frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda(t) P_0(t) + \frac{o(\Delta t)}{\Delta t} + \left(\frac{\mu}{k} \right) P_1(t) \quad (24)$$

So,

$$P'_0(t) = -\lambda(t) P_0(t) + \left(\frac{\mu}{k} \right) P_1(t) \quad (25)$$

At the specific operational peak t_{max} where traffic intensity reaches λ_{max} , the derivative of the empty state probability stabilizes to zero

($P'_0(t_{max}) = 0$):

$$P'_0(t_{max}) = -\lambda(t_{max}) P_0(t_{max}) + \left(\frac{\mu}{k} \right) P_1(t_{max}) = 0 \quad (26)$$

Definitions and Differential Difference Formulations for the Queueing Model

For a general system state n , there are two cases:

- **Case A: For System Size within service channel capacity** ($1 \leq n < c$):

$$\begin{aligned} P_n(t + \Delta t) = & P_n(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{1 - n \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n+1}(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{(n+1) \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n-1}(t) \{\lambda(t)\Delta t + o(\Delta t)\} \left\{1 - (n-1) \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \quad (27) \end{aligned}$$

- **Case B: For System Size exceeding service channel capacity** ($n \geq c$):

$$\begin{aligned} P_n(t + \Delta t) = & P_n(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{1 - c \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n+1}(t) \{1 - \lambda(t)\Delta t + o(\Delta t)\} \left\{c \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \\ & + P_{n-1}(t) \{\lambda(t)\Delta t + o(\Delta t)\} \left\{1 - c^* \left(\frac{\mu}{k}\right) \Delta t + o(\Delta t)\right\} \quad (28) \end{aligned}$$

where $c^* = c$ if $n > c$, and $c^* = c - 1$ if $n = c$.

Definitions and Differential Difference Formulations for the Queueing Model

So, we finally get,

$$\bar{P}_n = \frac{\bar{\lambda}^n}{n! \left(\frac{\mu}{k}\right)^n} \bar{P}_0, \quad 1 \leq n < c \quad (29)$$

$$\bar{P}_n = \frac{\bar{\lambda}^n}{c^{n-c} \left(\frac{\mu}{k}\right)^n c!} \bar{P}_0, \quad n \geq c \quad (30)$$

Similarly,

$$P_n = \frac{\lambda^n}{c^{n-c} \left(\frac{\mu}{k}\right)^n c!} P_0, \quad n \geq c \quad (31)$$

As, $\sum_{n=0}^{\infty} \bar{P}_n = 1$ and

$$\sum_{n=0}^{c-1} \bar{P}_n + \sum_{n=c}^{\infty} \bar{P}_n = 1 \quad (32)$$

Let $\bar{\rho}$ denote the traffic intensity ratio tracking parameter during peak operating hours, defined as $\bar{\rho} = \frac{\bar{\lambda}}{c\left(\frac{\mu}{k}\right)} = \frac{k\bar{\lambda}}{c\mu}$,

Definitions and Differential Difference Formulations for the Queueing Model

Then,

$$\bar{P}_0 \left[\sum_{n=0}^{c-1} \frac{(c\bar{\rho})^n}{n!} + \frac{c^c}{c!} \sum_{n=c}^{\infty} \bar{\rho}^n \right] = 1 \quad (33)$$

$$\bar{P}_0 = \frac{1}{\sum_{n=0}^{c-1} \frac{(c\bar{\rho})^n}{n!} + \frac{c^c}{c!} \{\bar{\rho}^c + \bar{\rho}^{c+1} + \dots\}} \quad (34)$$

provided that $\bar{\rho} < 1$,

$$\bar{P}_0 = \frac{1}{\sum_{n=0}^{c-1} \frac{(c\bar{\rho})^n}{n!} + \frac{c^c}{c!} \frac{\bar{\rho}^c}{1-\bar{\rho}}} \quad (35)$$

For $c = 2$,

$$\bar{P}_0 = \frac{1}{1 + c\bar{\rho} + \frac{2^2}{2!} \frac{\bar{\rho}^2}{1-\bar{\rho}}} = \frac{1}{1 + 2\bar{\rho} + \frac{2\bar{\rho}^2}{1-\bar{\rho}}} \quad (36)$$

$$\bar{P}_0 = \frac{1}{\frac{(1-\bar{\rho}) + 2\bar{\rho}(1-\bar{\rho}) + 2\bar{\rho}^2}{1-\bar{\rho}}} = \frac{1-\bar{\rho}}{1-\bar{\rho} + 2\bar{\rho} - 2\bar{\rho}^2 + 2\bar{\rho}^2} = \frac{1-\bar{\rho}}{1+\bar{\rho}} \quad (37)$$

Definitions and Differential Difference Formulations for the Queueing Model

Similarly,

$$\underline{P}_0 = \frac{1}{\sum_{n=0}^{c-1} \frac{(c\rho)^n}{n!} + \frac{c^c}{c!} \frac{(\rho)^c}{1-\rho}} \quad (38)$$

For $c = 2$,

$$\underline{P}_0 = \frac{1 - \rho}{1 + \rho} \quad (39)$$

Derivation of Mean Queue Lengths ($E(M)$)

Let, M = no. of persons in the queue when a new patient arrives at the hospital.

$$M = \begin{cases} 0 & \text{if } N \leq c \\ N - c & \text{if } N > c \end{cases} \quad (40)$$

Where, N = no. of persons in the hospital OPD.

Now, $E(M)$ = Mean queue length combining all the queues

Clearly, $E(M) = \sum_{n=c}^{\infty} (n - c) P_n$

Hence,

$$E(\bar{M}) = \sum_{n=c}^{\infty} (n - c) \bar{P}_n = \sum_{n=c}^{\infty} (n - c) \frac{\bar{\rho}^n}{c!} c^c \bar{P}_0 \quad (41)$$

$$E(\bar{M}) = \frac{c^c}{c!} \bar{P}_0 \sum_{n=c}^{\infty} (n - c) \bar{\rho}^n = \frac{c^c \bar{P}_0 \bar{\rho}^{c+1}}{c!} \sum_{n=c}^{\infty} (n - c) \bar{\rho}^{n-c-1} \quad (42)$$

$$E(\bar{M}) = \frac{c^c \bar{P}_0 \bar{\rho}^{c+1}}{c!} \sum_{u=0}^{\infty} u \bar{\rho}^{u-1} = \frac{c^c \bar{P}_0}{c!} [1 + 2\bar{\rho} + 3\bar{\rho}^2 + \dots] \bar{\rho}^{c+1} \quad (43)$$

Derivation of Mean Queue Lengths ($E(M)$)

$$E(\overline{M}) = \frac{c^c \overline{P}_0}{c!} \frac{1}{(1 - \overline{\rho})^2} (\overline{\rho})^{c+1} \quad (44)$$

For $c=2$,

$$E(\overline{M}) = \frac{2^2 \overline{P}_0}{2!} \frac{\overline{\rho}^{2+1}}{(1 - \overline{\rho})^2} = 2 \overline{P}_0 \frac{\overline{\rho}^3}{(1 - \overline{\rho})^2} \quad (45)$$

Similarly,

$$E(\underline{M}) = 2 \underline{P}_0 \frac{(\underline{\rho})^3}{(1 - \underline{\rho})^2} \quad (46)$$

These are the mean queue length combining all the c queues in the busy and slack hours respectively.

Derivation of Mean System Length ($E(N)$)

Now,

$$E(\bar{N}) = \sum_{n=0}^{\infty} n \bar{P}_n = \sum_{n=0}^{c-1} n \bar{P}_n + \sum_{n=c}^{\infty} n \bar{P}_n \quad (47)$$

$$E(\bar{N}) = \sum_{n=0}^{c-1} n \frac{(c\bar{\rho})^n}{n!} \bar{P}_0 + \sum_{n=c}^{\infty} n \frac{c^c (\bar{\rho})^n \bar{P}_0}{c!} \quad (48)$$

$$E(\bar{N}) = \sum_{n=1}^{c-1} \frac{n(c\bar{\rho})^n}{n!} \bar{P}_0 + \frac{c^c \bar{P}_0}{c!} \sum_{n=c}^{\infty} ((n-c) + c) \bar{\rho}^n \quad (49)$$

$$E(\bar{N}) = \sum_{n=1}^{c-1} \frac{(c\bar{\rho})^n}{(n-1)!} \bar{P}_0 + \frac{c^c \bar{P}_0}{c!} \left\{ \bar{\rho}^{c+1} \sum_{u=0}^{\infty} u \bar{\rho}^{u-1} + c \bar{\rho}^c \sum_{u=0}^{\infty} \bar{\rho}^u \right\} \quad (50)$$

$$E(\bar{N}) = \sum_{n=1}^{c-1} \frac{(c\bar{\rho})^n}{(n-1)!} \bar{P}_0 + \frac{c^c \bar{P}_0}{c!} \left\{ \bar{\rho}^{c+1} \frac{1}{(1-\bar{\rho})^2} + \frac{c \bar{\rho}^c}{(1-\bar{\rho})} \right\} \quad (51)$$

$E(\bar{N})$ = Mean no. of persons in the hospital OPD at busy hours

Similarly, $E(\underline{N})$, is obtained directly by substituting the parameters $\bar{\rho} \rightarrow \underline{\rho}$ and $\bar{P}_0 \rightarrow \underline{P}_0$ in the structural formula above.

Derivation of Mean Waiting Time In a Queue $E(T_q)$

$$P[\text{An arriving patient has to wait for purchasing ticket}] = \sum_{n=c}^{\infty} P_n \quad (52)$$

This probability evaluates to $\sum_{n=c}^{\infty} \bar{P}_n$ during busy blocks and $\sum_{n=c}^{\infty} \underline{P}_n$ during slack hours.

Let, T_q = Waiting time in a queue of an arriving patient

For $t > 0$,

$$F_{T_q}(t) = P[T_q \leq t] = P[T_q = 0] + P[0 < T_q \leq t] \quad (53)$$

$$F_{T_q}(t) = \sum_{n=0}^{c-1} P_n + \sum_{n=c}^{\infty} P_n P[0 < T_q \leq t \mid N = n] \quad (54)$$

$$F_{T_q}(t) = \sum_{n=0}^{c-1} P_n + \sum_{n=c}^{\infty} P_n \int_0^t \frac{(c(\frac{\mu}{k}))^n}{\Gamma(n)} e^{-c(\frac{\mu}{k})z} z^{n-1} dz \quad (55)$$

$$F_{T_q}(t) = \sum_{n=0}^{c-1} P_n + \sum_{n=c}^{\infty} \frac{c^c}{c!} \rho^n P_0 I_n(c\mu t) \quad (56)$$

Derivation of Mean Waiting Time IN a Queue $E(T_q)$

For $t > 0$,

Let $\omega = c \left(\frac{\mu}{k} \right) z$, which gives $d\omega = c \left(\frac{\mu}{k} \right) dz$. The internal integral transformation simplifies as:

$$f_{T_q}(t) = \sum_{n=c}^{\infty} \frac{c^c}{c!} \rho^n \frac{d \left(c \left(\frac{\mu}{k} \right) t \right)}{dt} \cdot \frac{d}{d \left(c \left(\frac{\mu}{k} \right) t \right)} \int_0^{c \left(\frac{\mu}{k} \right) t} \frac{e^{-\omega} \omega^{n-1}}{\Gamma(n)} d\omega \quad (57)$$

$$f_{T_q}(t) = \sum_{n=c}^{\infty} \frac{c^c}{c!} \rho^n \frac{c \left(\frac{\mu}{k} \right)}{\Gamma(n)} e^{-c \left(\frac{\mu}{k} \right) t} \left(c \left(\frac{\mu}{k} \right) t \right)^{n-1} \quad (58)$$

$$f_{T_q}(t) = e^{-c \left(\frac{\mu}{k} \right) t} \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \sum_{n=c}^{\infty} \frac{\left(c \left(\frac{\mu}{k} \right) t \rho \right)^{n-1}}{(n-1)!} \quad (59)$$

$$f_{T_q}(t) = e^{-c \left(\frac{\mu}{k} \right) t} \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[e^{c \left(\frac{\mu}{k} \right) t \rho} - \sum_{n=0}^{c-2} \frac{\left(c \left(\frac{\mu}{k} \right) t \rho \right)^n}{n!} \right] \quad (60)$$

Derivation of Mean Waiting Time In a Queue $E(T_q)$

Now, $E(T_q)$ = Mean waiting time of a newly arriving patient in a queue

$$E(T_q) = \int_0^{\infty} t \cdot f_{T_q}(t) dt$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\int_0^{\infty} t \cdot e^{-(1-\rho)c(\frac{\mu}{k})t} dt - \sum_{n=0}^{c-2} \frac{1}{n!} \int_0^{\infty} t \cdot e^{-c(\frac{\mu}{k})t} \left(c \left(\frac{\mu}{k} \right) t \rho \right)^n dt \right] \quad (61)$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \sum_{n=0}^{c-2} \frac{1}{n!} \left(c \left(\frac{\mu}{k} \right) \rho \right)^n \frac{(n+1)!}{\left(c \left(\frac{\mu}{k} \right) \right)^{n+2}} \right] \quad (62)$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \sum_{n=0}^{c-2} \frac{\rho^n (n+1)}{c^2 \left(\frac{\mu}{k} \right)^2} \right] \quad (63)$$

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{k} \right)^2} \left\{ \rho \sum_{n=1}^{c-2} n \rho^{n-1} + \sum_{n=0}^{c-2} \rho^n \right\} \right] \quad (64)$$

Derivation of Mean Waiting Time In a Queue $E(T_q)$

Then,

$$E(T_q) = \frac{c^c}{c!} \rho \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\rho)^2 c^2 \left(\frac{\mu}{k} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{k} \right)^2} \left\{ \frac{1-\rho^{c-1}}{1-\rho} + \frac{\rho(1-\rho^{c-2})}{(1-\rho)^2} - \frac{(c-2)\rho^{c-1}}{1-\rho} \right\} \right] \quad (65)$$

Using this we can derive the Expected Mean Waiting Time for the busy hours as well as the slack hours.

$$E(\overline{T_q}) = \frac{c^c}{c!} \bar{\rho} \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\bar{\rho})^2 c^2 \left(\frac{\mu}{k} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{k} \right)^2} \left\{ \frac{1-\bar{\rho}^{c-1}}{1-\bar{\rho}} + \frac{\bar{\rho}(1-\bar{\rho}^{c-2})}{(1-\bar{\rho})^2} - \frac{(c-2)\bar{\rho}^{c-1}}{1-\bar{\rho}} \right\} \right] \quad (66)$$

$$E(\underline{T_q}) = \frac{c^c}{c!} \underline{\rho} \left(\frac{\mu}{k} \right) \left[\frac{1}{(1-\underline{\rho})^2 c^2 \left(\frac{\mu}{k} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{k} \right)^2} \left\{ \frac{1-\underline{\rho}^{c-1}}{1-\underline{\rho}} + \frac{\underline{\rho}(1-\underline{\rho}^{c-2})}{(1-\underline{\rho})^2} - \frac{(c-2)\underline{\rho}^{c-1}}{1-\underline{\rho}} \right\} \right] \quad (67)$$

Derivation of Mean Waiting Time In a System $E(W)$

Now, we calculate mean waiting time of an arriving patient in the entire system of Hospital OPD before purchasing tickets, Then,

$$E(W) = E(T_q) + \text{Mean Service Time} \quad (68)$$

$$E(W) = E(T_q) + E\left(\frac{1}{\mu_n}\right) \quad (69)$$

Case-1: Approximate the second term as,

$$E\left(\frac{1}{\mu_n}\right) \approx \frac{k}{c\mu} \quad (70)$$

Then,

$$E(W) = E(T_q) + \frac{k}{c\mu} \quad (71)$$

Derivation of Mean Waiting Time In a System $E(W)$

Case-2: Here,

$$E\left(\frac{1}{\mu_n}\right) = \sum_{n=1}^{\infty} \frac{1}{\mu_n} P_n \quad (72)$$

Now we can derived it as,

$$\begin{aligned} E\left(\frac{1}{\mu_n}\right) &= \frac{K}{\mu} \sum_{n=1}^{c-1} \frac{1}{n} P_n + \frac{K}{c\mu} \sum_{n=c}^{\infty} P_n \\ &= \frac{K}{\mu} \sum_{n=1}^{c-1} \frac{P_n}{n} + \frac{K}{c\mu} \left(1 - \sum_{n=0}^{c-1} P_n\right) \\ &= \frac{K}{\mu} \frac{\lambda^n}{n! \left(\frac{\mu}{K}\right)^n} \cdot P_0 + \frac{K}{c\mu} \left(1 - \sum_{n=0}^{c-1} \frac{\lambda^n}{n! \left(\frac{\mu}{K}\right)^n} P_0\right) \\ &= \frac{K^2}{\mu^2} \frac{\lambda^n}{n!} \cdot P_0 + \frac{K}{c\mu} \left(1 - \sum_{n=0}^{c-1} \frac{\lambda^n}{n!} \left(\frac{K}{\mu}\right)^n P_0\right) \end{aligned}$$

Derivation of Mean Waiting Time In a System $E(W)$

$$\begin{aligned}E\left(\frac{1}{\mu_n}\right) &= \frac{K^2}{\mu^2} \lambda P_0 + \frac{K}{c\mu} \left(1 - P_0 - \lambda \frac{K}{\mu} P_0\right) \\&= \frac{\lambda K^2}{\mu^2} P_0 \left(1 - \frac{1}{c}\right) + \frac{K}{c\mu} (1 - P_0)\end{aligned}$$

$$\begin{aligned}\text{Now, } E\left(\frac{1}{\mu_n}\right) &= \lambda \cdot \left(\frac{K}{\mu}\right)^2 \cdot P_0 \left(1 - \frac{1}{c}\right) + \frac{K}{c\mu} (1 - P_0) \\&= \frac{\lambda K^2}{\mu^2} \frac{1 - \rho}{1 + \rho} \left(1 - \frac{1}{c}\right) + \frac{K}{c\mu} \left(1 - \left(\frac{1 - \rho}{1 + \rho}\right)\right)\end{aligned}$$

Case-3:

By Little's Formula we can write it as,

$$E(W) = \frac{E(N)}{\lambda} \quad (73)$$

Numerical Output of Our Queueing Model

Given parameters for our dataset:

$$\mu/K = 1.42, \quad \bar{\lambda} = 1.5, \quad \underline{\lambda} = 0.2, \quad c = 2$$

The traffic intensity for busy and slack hours respectively,

$$\bar{\rho} = \frac{\bar{\lambda} \cdot K}{2 \cdot \mu} = 0.58, \quad \underline{\rho} = \frac{\underline{\lambda} \cdot K}{2 \cdot \mu} = 0.07$$

For $c = 2$:

$$\bar{P}_0 = \frac{1 - \bar{\rho}}{1 + \bar{\rho}} = \frac{1 - 0.58}{1 + 0.58} = \frac{0.42}{1.58} = 0.2658 \approx 0.27$$

$$\underline{P}_0 = \frac{1 - \underline{\rho}}{1 + \underline{\rho}} = \frac{1 - 0.07}{1 + 0.07} = \frac{0.93}{1.07} \approx 0.87$$

Numerical Output of Our Queueing Model

Next, $E(M)$ = Mean Queue Length

For busy hours,

$$E(\overline{M}) = \frac{2 \cdot \overline{P}_0 \cdot \overline{\rho}^3}{(1 - \overline{\rho})^2} = \frac{2 \times 0.27 \times (0.58)^3}{(1 - 0.58)^2} = \frac{0.10536}{0.1764} = 0.5972 \approx 0.597$$

For slack hours,

$$E(\underline{M}) = \frac{2 \cdot \underline{P}_0 \cdot \underline{\rho}^3}{(1 - \underline{\rho})^2} = \frac{2 \times 0.87 \times (0.07)^3}{(1 - 0.07)^2} = \frac{0.008526}{0.8649} = 0.0098$$

The average queue length is 0.60 patients during busy hours and 0.01 patients during slack hours.

Numerical Output of Our Queueing Model

Then, $E(N)$ = mean numbers of persons in the hospital OPD

$$E(N) = \sum_{n=1}^{c-1} \frac{(c\rho)^n P_0}{n!} + \frac{c^c P_0}{c!} \left\{ \frac{\rho^{c+1}}{(1-\rho)^2} + \frac{c\rho^c}{1-\rho} \right\}$$

In our case $c = 2$, so $n = 1$:

For busy hours,

$$\begin{aligned} E(\bar{N}) &= \frac{2 \times 0.58 \times 0.27}{1} + \frac{4 \times 0.27}{2} \left\{ \frac{(0.58)^3}{(0.42)^2} + \frac{2 \times (0.58)^2}{0.42} \right\} \\ &= 0.3132 + 1.4623 = 1.7755 \end{aligned}$$

For slack hours,

$$\begin{aligned} E(\underline{N}) &= \frac{2 \times 0.07}{1} \times 0.87 + \frac{2^2 \times 0.87}{2!} \left\{ \frac{(0.07)^3}{(1-0.07)^2} + \frac{2 \times (0.07)^2}{1-0.07} \right\} \\ &= 0.1218 + 0.01902 = 0.14082 \end{aligned}$$

The average number of patients present in the OPD system is 1.78 during busy hours and 0.14 during slack hours.

Numerical Output of Our Queueing Model

Now, $E(T_q)$ = Expected Waiting Time in Queue

$$E(\overline{T}_q) = \frac{c^c}{c!} \bar{\rho}^c \left(\frac{\mu}{K} \right) \left[\frac{1}{(1 - \bar{\rho})^2 c^2 \left(\frac{\mu}{K} \right)^2} - \frac{1}{c^2 \left(\frac{\mu}{K} \right)^2} \right] \left\{ \frac{1 - \bar{\rho}^{c-1}}{1 - \bar{\rho}} + \frac{\bar{\rho}(1 - \bar{\rho}^{c-2})}{(1 - \bar{\rho})^2} - \frac{(c - 2)\bar{\rho}^c}{1 - \bar{\rho}} \right\}$$

Evaluating for $c = 2$:

For busy hours,

$$\begin{aligned} E(\overline{T}_q) &= \frac{6.5888}{2} \left[\frac{1}{0.1764 \times 2.0164} - \frac{1}{8.0656} \right] [1 + 3.2879 - 0] \\ &= 3.2944 \left(\frac{1}{0.35569} - \frac{1}{8.0656} \right) \\ &= 3.2944(2.8114 - 0.1239) \\ &= 3.2944 \times 0.5789 \\ &= 1.9071 \end{aligned}$$

Numerical Output of Our Queueing Model

For slack hours,

$$\begin{aligned} E(\underline{T_q}) &= 0.3976 \left(\frac{1}{7.501008 \times 8.0656} - \frac{1}{8.0656} \right) (1 + 0 - 0) \\ &= 0.3976 \times (0.13331 - 0.12398) \\ &= 0.0037 \end{aligned}$$

During busy hours, patients wait about 2 minutes on average before purchasing a ticket. During slack hours, the waiting time is almost negligible, meaning patients are served nearly immediately.

Numerical Output of Our Queueing Model

Now, we are estimating total system waiting time by 3 ways,

Case-1:

For busy hours,

$$E(\overline{W}) = E(\overline{T}_q) + \frac{k}{c\mu} = 1.9071 + 0.351 = 2.258 \quad (74)$$

For Slack hours,

$$E(\underline{W}) = E(\underline{T}_q) + \frac{k}{c\mu} = 0.0037 + 0.351 = 0.351 \quad (75)$$

On average, a patient spends about 2.26 minutes in the OPD system during busy hours and only 0.35 minutes during slack hours. This shows that the ticketing system is efficient and patients are served quickly throughout the day.

Numerical Output of Our Queueing Model

Case-2:

Now to calculate,

$$\begin{aligned}E\left(\frac{1}{\mu_n}\right) &= \left(\frac{1}{1.42}\right)^2 \times 1.5 \times 0.27 \times 0.5 + \frac{1}{1.42 \times 2}(1 - 0.27) \\&= 0.101 + 0.257 \\&= 0.358\end{aligned}$$

$$\begin{aligned}E\left(\frac{1}{\mu_n}\right) &= \left(\frac{1}{1.42}\right)^2 \times 0.2 \times 0.87 \times 0.5 + \frac{1 - 0.87}{1.42 \times 2} \\&= 0.586 + 0.0457 \\&= 0.632\end{aligned}$$

For busy hours,

$$E(\overline{W}) = 1.9071 + 0.358 = 2.26$$

For slack hours,

$$E(\underline{W}) = 0.0037 + 0.632 = 0.635$$

During busy hours, a patient spends about 2.26 minutes in the OPD system. During slack hours, the time reduces to only 0.63 minutes.

Numerical Output of Our Queueing Model

Case-3: Now by Little's formula

$$E(W) = \frac{E(N)}{\lambda}$$

For busy hours,

$$E(\overline{W}) = \frac{1.775}{1.5} = 1.133$$

For slack hours,

$$E(\underline{W}) = \frac{0.1408}{0.2} = 0.70$$

Using Little's Formula, the average time spent by a patient in the OPD system is about 1.13 minutes during busy hours and 0.70 minutes during slack hours.

Conclusion

- The queueing model shows that the hospital OPD system works efficiently and remains stable during both busy and slack hours.
- During busy hours, more patients arrive at the OPD, which increases queue length, waiting time and patient congestion.
- The waiting time in the queue is higher during peak hours, so patients may face moderate delays before getting service.
- The average number of patients in the system is also higher during busy periods, showing that the OPD becomes more crowded at that time.
- During slack hours, queue length and waiting time are very small, which means patients receive service quickly with almost no congestion.
- The study suggests that there is no requirement of any extra service counter or improving service speed as two counters are sufficient.

Overall, the queueing model helps to understand patient flow and service efficiency in the hospital OPD system and can support better hospital management decisions.

Thank You!

All R codes are available from the following GitHub repository:
<https://github.com/STAT-1092/STAT1092-Project-Research-2026>

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